

Worked Solutions

Discrete Mathematical Structures

Set A: Ensemble | CSE Semester 3

Companion to dms-ensemble.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Complement conditions

Correct option: (B). A complement joins with the element to give 1 and meets with it to give 0.

2. Complement uniqueness

Correct option: (D). Complement existence is a Boolean-algebra property, and distributivity gives uniqueness.

3. Boolean simplification

Correct option: (A). Distribution gives $(a \vee \neg a) \wedge (a \vee b) = 1 \wedge (a \vee b) = a \vee b$.

4. Disjunctive normal form

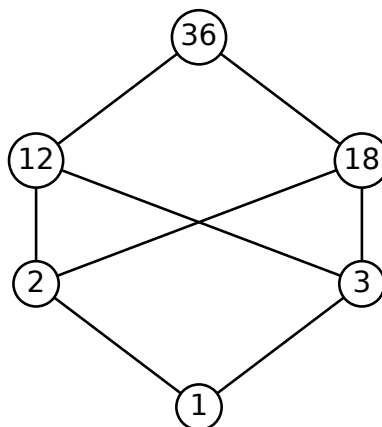
Correct option: (C). $(x \wedge y) \vee (\neg x \wedge z)$ is an OR of two AND terms of literals.

5. Conjunctive normal form

Correct option: (B). $(x \vee y) \wedge (\neg x \vee z)$ is an AND of two OR clauses of literals.

6. Divisibility poset and antichains

The cover relations are $(1, 2), (1, 3), (2, 12), (2, 18), (3, 12), (3, 18), (12, 36), (18, 36)$. They determine the Hasse diagram below; line crossings without a labelled vertex are not additional elements.



This poset is **not a lattice**. The common upper bounds of 2 and 3 are 12, 18 and 36. The two minimal common upper bounds 12 and 18 are incomparable, so there is no least upper bound. Equivalently, 12 and 18 have no greatest lower bound in the set: 2 and 3 are incomparable maximal common lower bounds. It is not valid to import gcd 6 into the set, since 6 is absent.

The complete list of antichains with more than one element is $\{2, 3\}, \{12, 18\}$. Every other distinct pair is comparable, so there are no larger antichains.

7. Bounded item selection

Let $u = A - 1$, $v = B - 3$, $w = C - 5$. Then $u + v + w = 6$, with $0 \leq u \leq 3$, $0 \leq v \leq 4$, $0 \leq w \leq 5$. The unrestricted nonnegative count is $\binom{8}{2} = 28$.

Subtract violations: $u \geq 4$ contributes $\binom{4}{2} = 6$, $v \geq 5$ contributes $\binom{3}{2} = 3$, and $w \geq 6$ contributes $\binom{2}{2} = 1$. Intersections are impossible because their minimum sums exceed 6. Thus

$$28 - 6 - 3 - 1 = 18.$$

Items of the same type are indistinguishable, so this counts feasible triples of quantities, not arrangements.

8. Partition bijection

For positive n , let λ be a partition of n with every part at least 3. In its Ferrers diagram, the first three columns have the same height k . Its conjugate partition $\mu = \lambda'$ therefore satisfies $\mu_1 = \mu_2 = \mu_3 = k$. Define a new partition

$$\nu = (\mu_1 + 2, \mu_2 + 1, \mu_3, \mu_4, \dots).$$

Its total is $n + 3$, and its three largest parts are $k + 2, k + 1, k$, which are consecutive.

Conversely, if a partition of $n + 3$ has largest parts $k + 2, k + 1, k$, subtract 2 from the first part and 1 from the second. The resulting partition has three equal largest parts k and total n . Conjugating it gives a partition of n in which every row has at least three cells, hence every part is at least 3. These constructions undo each other, proving the claimed equality. For $n=1$ or 2 , both sets are empty.

9. The 75th Fike permutation

Use the original descending-exchange Fike convention, with the original arrangement 01234 counted as permutation 1. Enumerate (d_2, d_3, d_4, d_5) with $d_k = k, k - 1, \dots, 1$, and d_5 varying fastest. For each tuple, start from 01234 and exchange positions k and d_k in increasing k order, using one-based positions.

The zero-based index is $75 - 1 = 74$. The block weights for these four choices are 60, 20, 5 and 1:

$$74 = 1(60) + 0(20) + 2(5) + 4(1).$$

Thus $(d_2, d_3, d_4, d_5) = (2 - 1, 3 - 0, 4 - 2, 5 - 4) = (1, 3, 2, 1)$.

Exchange	Arrangement
Start	01234
Positions 2 and 1	10234
Positions 3 and 3	10234
Positions 4 and 2	13204
Positions 5 and 1	43201

Hence the answer under this convention is **43201**.

Convention reference: J. S. Rohl, "Programming Improvements to Fike's Algorithm for Generating Permutations," The Computer Journal 19(2), 156–159 (1976), Figure 1 and recursive procedure. <https://doi.org/10.1093/comjnl/19.2.156>. The convention is stated explicitly because Fike order is not lexicographic order.

10. Regular self-complementary graph

Assume a nonempty simple graph. If G is r -regular on n vertices, its complement is $(n - 1 - r)$ -regular. Isomorphism to its complement implies

$$r = n - 1 - r, \quad r = (n - 1)/2.$$

Therefore n is odd. The handshake lemma says $nr = 2|E|$ is even. Since n is odd, r must be even; write $r = 2t$. Substitution gives $n = 2r + 1 = \boxed{4t + 1}$, with $t \geq 0$.

11. Minimum degree forces connectivity

Suppose G is disconnected, and a component has s vertices. A vertex in that component has degree at most $s - 1$, so

$$s - 1 \geq \delta(G) \geq (p - 1)/2, \quad s \geq (p + 1)/2.$$

Every component must satisfy this bound. Two components would already contain at least $p+1$ vertices, contradicting the total p . Hence G is connected. This argument assumes the usual simple-graph setting; a one-vertex graph is connected directly.

12. Dijkstra on the directed source network

Respect arrow direction. The outgoing edges are: $v_1 \rightarrow v_2 : 4$, $v_1 \rightarrow v_3 : 5$; $v_2 \rightarrow v_1 : 6$, $v_2 \rightarrow v_3 : 4$, $v_2 \rightarrow v_5 : 8$; $v_3 \rightarrow v_2 : 6$, $v_3 \rightarrow v_4 : 2$, $v_3 \rightarrow v_6 : 3$; $v_4 \rightarrow v_5 : 7$; $v_5 \rightarrow v_1 : 3$, $v_5 \rightarrow v_4 : 2$; $v_6 \rightarrow v_3 : 5$, $v_6 \rightarrow v_4 : 4$. The table records tentative distances after relaxing outgoing edges of each settled vertex.

Settled	d_1	d_2	d_3	d_4	d_5	d_6
Initial	0	∞	∞	∞	∞	∞
v_1	0	4	5	∞	∞	∞
v_2	0	4	5	∞	12	∞
v_3	0	4	5	7	12	8
v_4	0	4	5	7	12	8
v_6	0	4	5	7	12	8
v_5	0	4	5	7	12	8

Destination	Shortest path	Length
v_1	v_1	0
v_2	$v_1 \rightarrow v_2$	4
v_3	$v_1 \rightarrow v_3$	5
v_4	$v_1 \rightarrow v_3 \rightarrow v_4$	7
v_5	$v_1 \rightarrow v_2 \rightarrow v_5$	12
v_6	$v_1 \rightarrow v_3 \rightarrow v_6$	8